Module 12: Final Project Workbench

Complete system, evaluation report, and demo

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Primary sources: Course synthesis from Osmani; Thomas & Hunt project practices.

Learning Outcomes

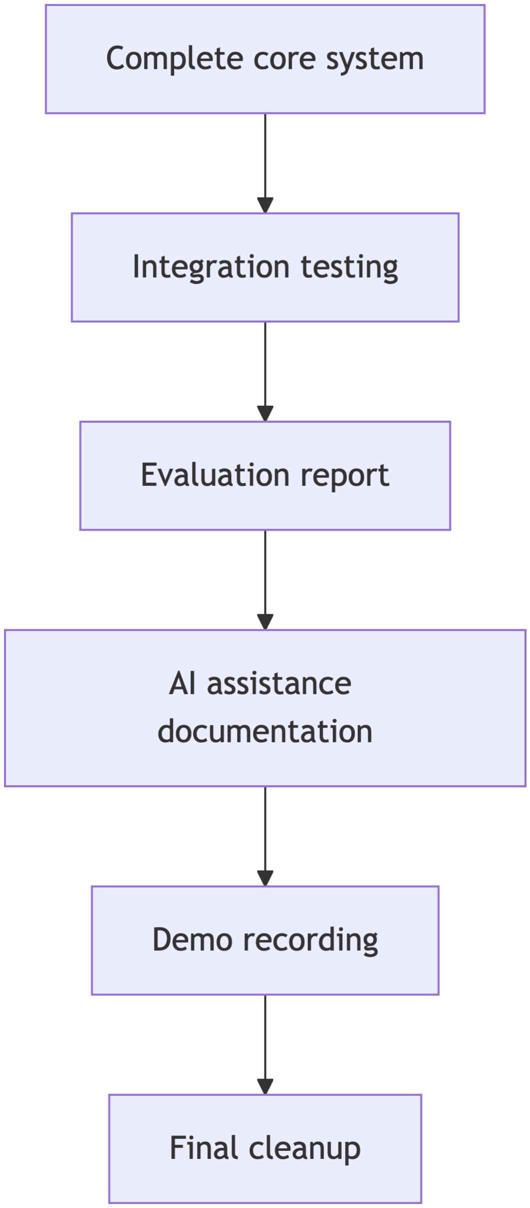
- Design and implement a complete AI-assisted system.
- Integrate workflows, evaluation, and safeguards.
- Justify architectural and engineering decisions.
- Present and defend the system.

Final System Expectations

Your Controlled Coding Agent should demonstrate:

- Structured input handling.
- AI-assisted generation.
- Evaluation and testing.
- Controlled workflow management.
- Persistence or artifact tracking.
- Documentation.
- Human oversight.
- Collaborative engineering practice.

Final Submission Timeline



Reproducibility

A grader should be able to answer:

- How do I set it up?
- How do I run it?
- What example should I try?
- What output should I expect?
- How do I run tests?
- Where are evaluation results?
- Where is AI assistance documented?

Evaluation Report

The evaluation report should include:

- What was evaluated.
- Metrics or criteria.
- Test cases or scenarios.
- Results.
- Interpretation.
- Known limitations.
- What would improve next.

AI Assistance Section

Your README should make AI use transparent:

- Planning.
- Code generation.
- Test generation.
- Debugging.
- Review.
- Integration support.
- Documentation support.
- Accepted, modified, and rejected suggestions.
- Verification methods.

Demo Structure

Recommended 5-8 minute flow:

1. Problem and users.
2. Architecture overview.
3. Live or recorded workflow run.
4. Evaluation results.
5. Safeguards and human oversight.
6. Team process and AI evidence.
7. Lessons learned.

Professional Defense

Be ready to explain:

- Why this architecture?
- What did AI help with?
- What did humans correct?
- What risks remain?
- Which tests matter most?
- What would you do next with more time?

Final Quality Checklist

- Clean repository organization.
- No committed secrets.
- Fresh-environment setup tested.
- Tests and examples run.
- Demo link included.
- Required document paths submitted.
- Commit hash captured.
- Individual contributions clear.

Course Synthesis

Responsible AI-assisted engineering means:

- Clear intent.
- Structured prompts.
- Human review.
- Automated evidence.
- Risk awareness.
- Transparent documentation.
- Team accountability.

Closing Reflection

Answer individually:

- What AI interaction changed your process most?
- What risk did you underestimate early?
- What verification habit will you keep?
- What work should remain human-led?

Takeaways

- The final project is a demonstration of engineering judgment.
- AI assistance is strongest when controlled, evaluated, and documented.
- The goal is not to show that AI wrote code; it is to show that your team built trustworthy software with AI assistance.